Vulnerability Assessment Report

1st January 20XX

Scenario:

You are a newly hired cybersecurity analyst for an e-commerce company. The company stores information on a remote database server, since many of the employees work remotely from locations all around the world. Employees of the company regularly query, or request, data from the server to find potential customers. The database has been open to the public since the company's launch three years ago. As a cybersecurity professional, you recognize that keeping the database server open to the public is a serious vulnerability.

You are tasked with completing a vulnerability assessment of the situation to communicate the potential risks to decision makers at the company. You must create a written report that explains how the vulnerable server is a risk to business operations and how it can be secured.

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 2026 to August 2026. NIST SP 800-30 Rev. 1 is used to guide the risk analysis of the information system.

Purpose

The purpose of this vulnerability assessment is to evaluate the risks associated with the company's publicly accessible remote database server.

The database server is valuable to the business because it allows employees to access information remotely from different locations around the world. It is also an important tool for identifying potential customers, supporting the company's sales and business development activities.

It is important for the company to secure the information stored on the server because it contains private business and potentially sensitive customer information. If the database remains publicly accessible, malicious actors could exploit this vulnerability to gain unauthorised access, steal or manipulate data.

If the server were disabled or became unavailable, this could significantly disrupt business operations. Employees would lose access to an important resource used to identify potential customers, potentially affecting productivity, sales opportunities and the company's ability to operate effectively.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Competitor	Steal data to obtain potential employees	3	1	3
Hacker	Unauthorised access, allowing them to alter/delete critical information	2	3	6
Malicious Software	Disrupt mission-critical operations by making the database unavailable	2	3	6

Approach

The three threat sources selected were competitors, hackers and malicious software because each could exploit the vulnerabilities associated with the publicly accessible database server. Competitors could attempt to steal confidential customer and business information to gain a competitive advantage. Hackers could gain unauthorised access to the database and alter or delete important records. Malicious software could disrupt or disable the server, preventing employees from accessing information needed to identify potential customers. These threats

are significant because they could result in data loss, financial damage and disruption to day-to-day business operations.

Remediation Strategy

The company should restrict public access to the database and implement the principle of least privilege, ensuring employees can only access information required for their roles. Multi-factor authentication (MFA) should be introduced to reduce the risk of unauthorised access through compromised credentials. Implementing a defence-in-depth approach, including firewalls, anti-malware software and regular security updates, would help protect against external attacks and malicious software. The Authentication, Authorisation and Accounting (AAA) framework should also be used to verify users, control access permissions and monitor database activity.